Multiple Attention Network for *Spartina alterniflora* Segmentation Using Multitemporal Remote Sensing Images

Boyu Zhao, Mengmeng Zhang, Jianbu Wang, Xiukai Song, Yuanyuan Gui, Yuxiang Zhang, *Student Member, IEEE*, and Wei Li, *Senior Member, IEEE*

***Abstract*— The semantic segmentation of multitemporal remote sensing images to construct wetland land surface coverage is the basis for the perception and dynamic modeling of geographic scenes. However, the segmentation of Spartina alterniflora in remote sensing images on wetlands faces the problems such as a low level of cooperative interpretation in multitemporal images and high fragmentation in the distribution of *S. alterniflora*. To solve the issues, a multiple attention network (MARNet) based on transfer learning is proposed. The method is designed with a plug-and-play attention module to enhance the learning of vegetation features and improve the network's ability to focus on small areas of *S. alterniflora*. At the same time, MARNet designs the transfer learning architecture from both interdomain alignment and intradomain adaptation perspectives, aligning the statistical distribution by using the maximum mean difference (MMD) between the source and target domains (TDs), and entropy minimization within the domain of the TD to enhance the high confidence prediction of this domain. In addition, since the samples have a serious imbalance problem, redundant cutting and splicing steps are employed for the prediction results to prevent the poor edge prediction of some image blocks. Experimental results on three cross-year RSI datasets demonstrate that the proposed MARNet performs significantly better than other networks and is able to extract *S. alterniflora* in wetlands more accurately.**

***Index Terms*— Attention block, deep learning, multitemporal remote sensing images, *Spartina alterniflora* segmentation, transfer learning.**

NOMENCLATURE

Representation	Explanation
Feature map	*Width* × *height* × *channel*-dimensional tensor. The sizes of *width*, *height*, and *channel* are represented by W, H, and C, respectively.
$conv_{k\times k}$	Convolution operation with a $k \times k$ size kernel, the default value of k is 3.
$conv1D_{k\times k}$	One-dimensional convolution operation with a $k \times k$ size kernel, the default value of k is 5.
BN	Batch normalization operation, a basically normalization operation.
$ReLU$	Rectified linear unit, used as an activation function.
$mp(f,k\times k,s)$	Max pooling operation, works on the feature map f as a maximum filter with $k\times k$ kernel and s stride.
$ap(f,k\times k,s)$	Average pooling operation, works on the feature map f as an average filter with $k\times k$ kernel and s stride.
$up(f,k\times k,s)$	Up-sampling operation, use bilinear interpolation to expand the width and height of the feature map to double the original size, at this case the stride length s is 2 and the width of the kernel k is 2,
$GMaxPool$	Global max pooling operation in channel dimension. For feature map $f_{\text{in}} \in \mathbb{R}^{W\times H\times C}$, the operation of $GMaxPool$ yields $f_{\text{out}} \in \mathbb{R}^{W\times H\times 1}$.
$GAvgPool$	Global average pooling operation in channel dimension. For feature map $f_{\text{in}} \in \mathbb{R}^{W\times H\times C}$, the operation of $GAvgPool$ yields $f_{\text{out}} \in \mathbb{R}^{W\times H\times 1}$.
$concate$	The concatenate operation, for two feature maps $f_1 \in \mathbb{R}^{W\times H\times C_1}$, $f_2 \in \mathbb{R}^{W\times H\times C_2}$, the operation of $concate$ yields $[f_1, f_2] \in \mathbb{R}^{W\times H\times (C_1+C_2)}$.

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I. INTRODUCTION

REMOTE sensing techniques can obtain more accurate information on the spatial and temporal distributions of Spartina alterniflora (*S. alterniflora*) in wetlands compared with field surveys because of the advantages of large monitoring areas and long data time series [1], [2]. Early remote sensing detection of *S. alterniflora* in wetlands used traditional machine learning methods such as maximum likelihood, support vector machines, and random forests [3], [4]. For coastal wetlands with rich vegetation species and large ranges, the interpretation accuracy obtained using traditional methods is low and

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cannot accurately grasp the temporal dynamics of *S. alterniflora* invasion.

Owing to the limitations of traditional machine learning [5], [6], deep learning methods have been gradually applied to wetland vegetation classification [7], [8]. Deep learning-based methods can extract deep-level features more effectively and greatly improve the classification accuracy of features in complex environments [9], [10]. For example, Guo et al. [8] used a mangrove extraction network (ME-Net) to classify mangroves on Hainan Island, and the overall accuracy of mangrove extraction was 97.48%, Zhu et al. [11] used multiscale residual convolutional neural network (MRCNN) to classify *S. alterniflora* in coastal wetlands of Jiangsu Province. However, multispectral remote sensing data of wetlands include more complex land categories and higher processing efficiency requirements. This characteristic makes it more appropriate to perform wetland segmentation [12], [13].

U-shaped symmetric network (U-Net) has shown advantages in the field of image semantic segmentation [14], which can fuse low and high-dimensional features, greatly improving segmentation accuracy [15], [16]. However, U-Net is highly prone to semantic gaps due to its structure of directly splicing shallow features of the encoder and deep features of the decoder across the connection layer [17], which leads to the loss of smaller targets during segmentation [18]. Therefore, Oktay et al. [19] proposed U-Net structures with attention gates (AGs), and the proposed AGs improved the sensitivity and classification accuracy by suppressing features in irrelevant regions. Nevertheless, these methods failed to take into account the spectral and distribution characteristics of *S. alterniflora*, which has better type differentiation in the wavelength range of the near-infrared band [11], and the wetland ecosystem in which it is located has a complex land cover pattern and complex remote sensing imaging conditions [20].

Most of the existing deep learning-based segmentation methods [21] are only applicable to fixed scenarios, i.e., the training and test samples are independently and identically distributed. Compared with the semantic segmentation of remote sensing images in fixed time and the same scene, processing and analyzing multiple remote sensing images in the same area covered by different time [22], [23], is more conducive to the study of the invasion mechanism of *S. alterniflora* [24]. In addition, the existing methods cannot be transferred to other scenes or tested in real-time without labels. And the acquisition of multispectral remote sensing images in different years is inevitably affected by various factors, such as different seasons and different weather conditions [25], causing spectral reflectance differences between the training sample in the source domain (SD) and the test sample in the target domain (TD) of the same land cover class. Therefore, existing methods often encounter the problem of spectral shift when segmenting TD directly using SD. Therefore, it is particularly important to improve the approach based on transfer learning.

To address these issues, this study proposes a multiple attention network (MARNet) based on transfer learning using multispectral remote sensing images from Landsat 5 TM and Landsat 8 OLI satellites for multiple years, and the main contributions are summarized as follows.

1) MARNet is developed as a specialized model for segmenting *S. alterniflora* in wetlands, which aims to capture consistent representations across multitemporal remote sensing data. It extracts invariant features using multilevel feature aggregation and an attention mechanism (AM), reducing its dependence on multitemporal data.
2) Based on the spectral and distributional properties of *S. alterniflora*, a multiple attention (MA)-block is proposed. This block mitigates semantic gaps caused by directly concatenating shallow and deep features in traditional segmentation networks. Also, it enhances feature extraction using the interplay of multiple AMs. The designed attention structure is experimentally verified to achieve state-of-the-art performance.
3) To alleviate spectral drift and spatial distribution differences, a transfer learning-based MARNet network framework is constructed from two perspectives: interdomain alignment and intradomain adaptation, where interdomain operations are designed to align the statistical distribution between SD and TD, and intradomain operations are applied to improve the high confidence prediction of TD, and this combination helps to improve the network framework for *S. alterniflora* extraction in multitemporal and multispectral remote sensing images.

The remainder of this article is organized as follows. The related work is described in Section II. The proposed framework is introduced in Section III. The experimental results and analysis are presented in Section IV. The ablation experiments on the innovation sites are described in Section V. Finally, Section VI summarizes with concluding remarks.

II. Related Work

Some techniques used in the proposed method are described. Assume that $X_s = \{x_i^s\}_{i=1}^{n_s} \in \mathbb{R}^d$ and $X_t = \{x_i^t\}_{i=1}^{n_t} \in \mathbb{R}^d$ are data from SD and TD, respectively, and $Y_s = \{y_i^s\}_{i=1}^{n_s} \in \mathbb{R}^d$ are the corresponding class labels. Here, d, n_s, and n_t denote the dimension of data, the number of source samples, and the number of target samples, respectively.

A. Semantic Segmentation Based on Deep Learning

There are many popular frameworks for semantic segmentation, such as the classical semantic segmentation network U-Net [14] and Deeplabv3 [26]. Some semantic segmentation networks are widely recognized for their superior performance in certain tasks. High-resolution network (HRNet) [27] used a multibranch structure in a whole network, and the resolution of the feature map remained during the forward propagation in each branch. The three branches of HRNet were individual but information was exchanged between them. Pyramid scene parsing network (PSPNet) [28] employed pyramid scenario resolution to enhance the network's global scope information extraction capability. In recent years, ResUNet was proposed to extract roads [29], and it has achieved good results by using U-Net as a backbone while using residual connections. The pure convolutional neural network [30] (ConvNeXt) is optimized from the training strategy, using the deep separable convolution in Swin transformer, inverse bottleneck layer, large convolutional kernel, and other strategies to have better performance in the final.

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B. Attention Mechanism

The AM is now being incorporated into semantic segmentation networks [31], [32]. ResNeSt [33] added channelwise AM to ResNet, it may have weights on different channels and enhance message passing on different scales and features. Global context network (GCNet) [34] overcame the drawback of excessive computation of a nonlocal mechanism by proposing a global context AM. The criss-cross network (CCNet) [35] included a cross-cross-attention module that can be utilized to capture contextual information from remote dependencies more efficiently and effectively, enhancing the semantic understanding of the model.

In general, AM introduces an attention weight for the input and uses it to distinguish between auxiliary task information and relatively irrelevant information [36], [37], which is mainly learned by adding feedforward neural network structure to the architecture. For example, AGs [19], squeeze and excitation blocks (SE-blocks) [38], efficient channel attention modules (ECA) [39], convolutional block attention module (CBAM) [40], position attention module (PAM), and channel attention module (CAM) [41] are structures commonly used. In addition, interlaced sparse self-attention (ISA) [42], coordinate attention (CA) [43], efficient pyramid squeeze attention (EPSA) [44], position attention block (PAB), and channel attention block (CAB) [21] are more advanced attention modules.

AGs are presented in Attention-UNet [19], which employs U-Net with soft attention to supervising the feature map of the previous level by the feature map of the next level, restrict the activation to the region to be segmented, and reduce the activation value of the background to optimize the segmentation. ECA [39] optimizes the segmentation by not downscaling the local cross-channel interaction strategy, which effectively avoids the effect of dimensionality reduction on the channel attention learning effect, the module has less number of parameters but has significant effect gain. PAM and CAM are modeled by dual attention network (DANet) [41], which improves the feature representation of the semantic segmentation network by modeling semantic dependencies in spatial dimension and channel dimension, respectively, which helps to obtain more accurate segmentation results. ISA is proposed by interlaced sparse self-attention network (ISANet) [42], which greatly alleviates resource consumption by adjusting the distribution of elements and establishing local connections. CA [43] extracts more sensitive features in terms of orientation and location to enhance the representation of objects of interest. EPSA [44] extracts finer-grained multiscale spatial information to establish longer distance channel dependencies. PAB and CAB [21], on the other hand, improve on PAM and CAM to enhance the feature representation of segmented objects in space.

C. Maximum Mean Discrepancy

Among the existing interdomain alignment methods for transfer learning, maximum mean discrepancy (MMD) [45] is an effective nonparametric distance measure and is one of the most widely used strategies to measure the difference in distribution between SD and TD. The MMD distance of two different domains is described as

$$\mathrm{MMD}(X_s, X_t) = \left\| \frac{1}{n_s} \sum_{i=1}^{n_s} \phi(x_i^s) - \frac{1}{n_t} \sum_{j=1}^{n_t} \phi\left(x_j^t\right) \right\|_H^2 \quad (1)$$

$\phi(\cdot)$ represents a mapping function from the original space to reproducing kernel Hilbert space (RKHS), and H indicates that the MMD distance is measured by mapping the data into the RKHS.

The meaning of the formula is to calculate the mean discrepancy between SD and TD in RKHS. The smaller the value of MMD, the more similar the two domains, and vice versa. By square expansion of the formula, the inner product in RKHS space can be converted into a kernel function, so the MMD can be calculated directly through the kernel function.

D. Minimize Prediction Entropy

The entropy minimization algorithm constrains the model by minimizing the entropy of the prediction results, which in turn allows the model to produce high-confidence predictions and reduces the low-confidence predictions of the target class of images, i.e., high-entropy predictions. This method has been shown to work well for semi-supervised learning [46], [47] and clustering [48], [49], while the algorithm has also been gradually applied to in-domain adaptation for classification tasks [50], and [51] also successfully applied unsupervised training based on entropy loss to the task of semantic segmentation, enhancing the stability of the model during training by penalizing the high entropy prediction of the TD's images, while obtaining competitive performance.

III. Proposed MARNet Framework

MARNet is a transfer-learning-based semantic segmentation network for *S. alterniflora* segmentation. The overall network structure is illustrated in Fig. 1. MARNet is based on the standard encoder–decoder structure with the addition of an MA-block, a loss function based on transfer learning, and a post-processing component.

The MA-block aggregates the features of *S. alterniflora* from three different directions: contextual attention, channel attention, and spatial attention. The transfer-learning loss function uses two orientations, cross-domain alignment and intradomain adaptation, to optimize the segmentation of *S. alterniflora*. Specifically, patches from SD (with label) and patches from TD (without label) are put into MARNet to extract features with abundant semantic information. The features of SD and TD extracted by MARNet are denoted as Z_s and Z_t for the distribution alignment of SD and TD and the entropy minimization of TD, respectively. In addition to the overall structure, MARNet also uses a post-processing component to reduce the impact of edge prediction on the overall prediction results.

A. Multiattention Residual Network Framework

MARNet's network framework consists of four blocks with different functions, including input block, encoding block, feature fusion block, output block, and MA-block. The abbreviations of some important operations used by MARNet and their descriptions are listed in the Nomenclature.

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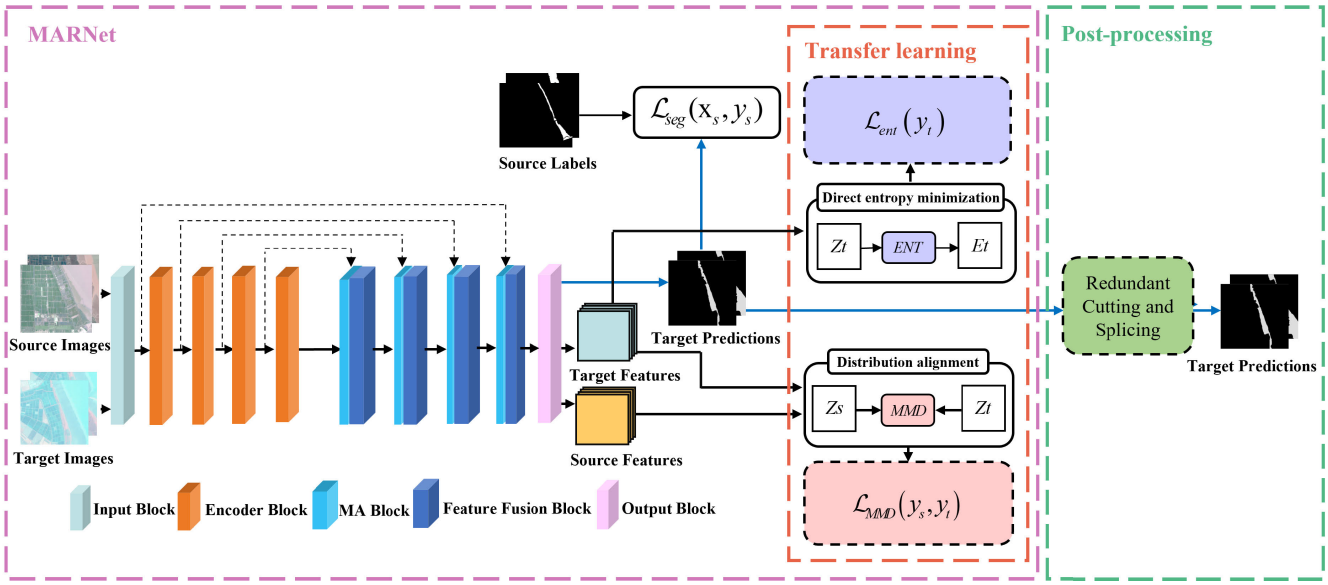


Fig. 1. Overall structure of MARNet for *S. alterniflora* segmentation in wetlands using Landsat multispectral remote sensing images. The figure includes the overall network framework with the addition of MA-blocks, the transfer learning part, and the post-processing part.

1) Input Block and Encoder Block: Both the input block and encoder block are used to extract common features from the multispectral image, and they are designed to learn about the characteristics of the land on which the *S. alterniflora* is located and other land types.

The input block consists of a residual connection and two convolution operations. Its function is to extract common features of the original images. When the channel number of features increases, the residual connection prevents the previous features from being lost [52]. The input to the input block is the multispectral image provided by LandSat5 and LandSat8, which is the original image. The input of the block is six-spectrum RSIs denoted as Img, while the output is a feature map denoted as x_i. The specific operation steps of the input block are

$$\begin{aligned} x_1 &= \mathrm{ReLU}(\mathrm{BN}(\mathrm{conv}_{3\times3}(\mathrm{Img}))) \\ x_2 &= \mathrm{ReLU}(\mathrm{BN}(\mathrm{conv}_{3\times3}(x_1))) \\ x_3 &= \mathrm{BN}(\mathrm{conv}_{1\times1}(\mathrm{Img})) \\ x_4 &= x_2 + x_3. \end{aligned} \tag{2}$$

The encoding block has an additional step of downsampling compared with the input block. The series of down-sampling operations in MARNet enable the network to extract features from images with different sizes. The max pooling operation is used as a down-sampling operation. x_E denotes the tensor being operated on the encoder block. The specific operation steps of the encoder block are

$$\begin{aligned} x_{E_1} &= \mathrm{mp}(x_E, 2\times2, 2) \\ x_{E_2} &= \mathrm{ReLU}(\mathrm{BN}(\mathrm{conv}_{3\times3}(x_{E_1}))) \\ x_{E_3} &= \mathrm{ReLU}(\mathrm{BN}(\mathrm{conv}_{3\times3}(x_{E_2}))) \\ x_{E_4} &= \mathrm{BN}(\mathrm{conv}_{1\times1}(x_{E_1})) \\ x_{E_5} &= x_{E_3} + x_{E_4}. \end{aligned} \tag{3}$$

2) Multiattention Block: The MA is located before the feature fusion block. It is designed to have three functions: obtaining hints on the location of the *S. alterniflora*, sorting out the most informative channels for *S. alterniflora* segmentation, and optimizing segmentation by guiding the feature response of the above information to reduce the irrelevant background region. The filtering of spatial dimensional information, channel dimensional information, and contextual information by the MA-block allows the feature fusion module to obtain features focused on the textural and spectral properties of *S. alterniflora*.

The structure of the MA is illustrated in Fig. 3. MA-block has two inputs, which are the feature map of the feature fusion part and the feature map of the encoder part of its upper layer. The fusion feature map is upsampled to obtain a feature map of the same size as the encoder map. The attentional feature map is obtained by adding this feature map with the corresponding elements of the encoder map. The attentional feature map contains contextual information, and because the contextual information possesses richer features through further convolution, the contextual attention of the above information can be directly suppressed by using the below information to direct the task-irrelevant feature responses, thus achieving the effect of highlighting the region encroached by *S. alterniflora*. This contextual attention paves the way for the subsequent channel attention branching operations and spatial attention branching operations for a feature highlighting.

The operation on the attentional feature map has two branches. The first branch is to find the channels from the attentional feature map that aids feature extraction. Because of the spectral properties of *S. alterniflora* [11] that allow it to be well distinguished in the near-infrared band, but poorly distinguished in other bands. To avoid the loss of channel information, the 1-D convolution-based channel interaction strategy in ECA-block [39] is used to produce a channel AM. Compared with SE-block [38], this local cross-channel interaction strategy can avoid the loss of information between channels due to dimensionality reduction, and also learn high-performance channel attention. The encoder feature map

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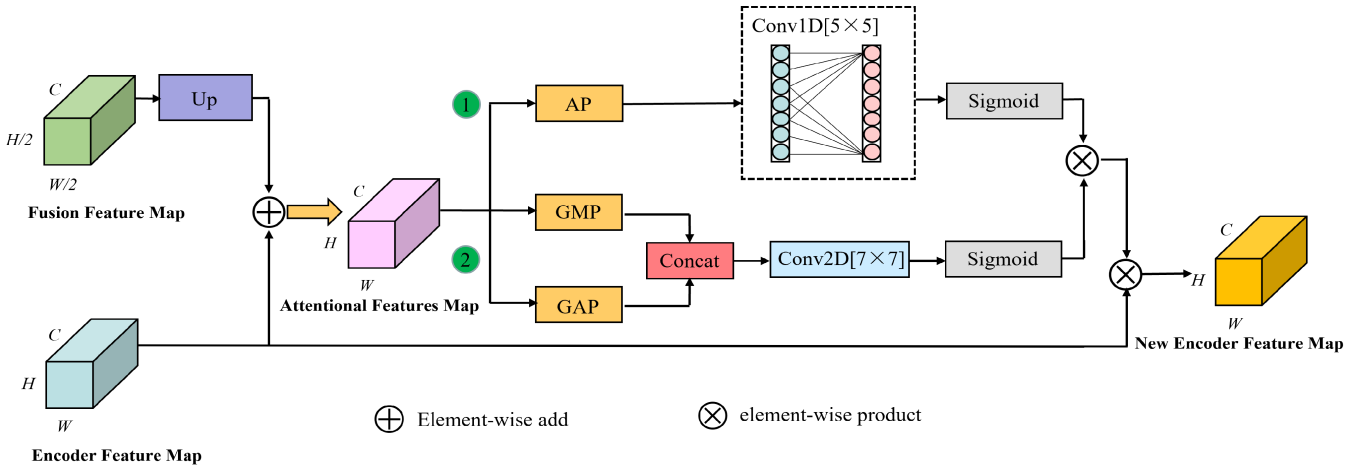


Fig. 2. Structure of MA-block. C represents channel number, W represents width, and H represents height. The encoder feature map is the output feature from the previous block, and the fused feature map is the feature extracted by the feature fused block of the corresponding size. The coarse arrows represent the aggregation of contextual information. Marker 1 represents the channel attention branch, and marker 2 represents the spatial attention branch.

is denoted as x_E, the fused feature map is denoted as x_F, the attentional feature map is denoted as x_A, and the weight coefficient of the output is defined as x_{w_1}. The operation of the first branch can be expressed as

$$\begin{aligned} x_A &= \mathrm{up}(x_F, 2\times 2, 2) + x_E \\ x_{w_1} &= \mathrm{Sigmoid}(\mathrm{conv1}D_{5\times 5}(\mathrm{ap}(x_A, W\times H, 1))). \end{aligned} \tag{4}$$

The function of the second branch is to obtain information on the location of *S. alterniflora*. Because *S. alterniflora*, as an invasive plant, occupies a very small area in the studied region, the number of samples differs significantly from the number of background samples. This distribution characteristic motivated the inclusion of this branch. The detailed sample points and their proportions are listed in Table I. This branch first performs GMaxPool and GAvgPool operations on the channel dimensions separately. Max pooling can filter out useless information and make the characteristics more distinct, while average pooling is toward overall characteristics, which can prevent the loss of information besides the extremum. Considering the distribution characteristics of *S. alterniflora*, both the maximum and average features must be retained. Then, the feature maps with 1-channel obtained by different pooling operations are combined, and finally, the final weight coefficients x_{w_2} are obtained by convolution operation and activation function. Multiplying the encoded feature map with x_{w_1} and x_{w_2} yields a new feature map for the jump-over connection in the feature fusion module. These operations are expressed as

$$\begin{aligned} &\mathrm{AP} = \mathrm{GAvgPool}(x_A), \quad \mathrm{MP} = \mathrm{GMaxPool}(x_A) \\ &x_{w_2} = \mathrm{Sigmoid}(\mathrm{conv}_{7\times 7}(\mathrm{concate}(AP, MP))) \\ &x_{\mathrm{new}} = x_{w_2} * x_{w_1} * x_E. \end{aligned} \tag{5}$$

The function of the feature fusion block is to join the feature map x_{new} obtained by the attention module with the feature map x_{inf} extracted by downsampling in the same layer while supplementing the feature details of this size with residual and convolution operations. This block splices the shallow features filtered by the attention module with the deep features and gradually restores the features to the size of the original image by the up-sampling operation to achieve the segmentation of grass in wetlands. The tensor generated in the feature fusion process is denoted by x_F, and the operations of the feature fusion block are expressed as

TABLE I

PROPORTION OF SAMPLE POINTS FOR *S. Alterniflora* OF DIFFERENT YEARS

Year	Total sample points	Sample points of S.alterniflora	Proportion
1995	15,995,311	47238	0.295%
2000	16,002,429	161444	1.008%
2005	16,423,106	179006	1.090%
2010	16,250,083	188225	1.158%
2015	14,899,939	191270	1.284%
2020	15,473,293	196280	1.269%

3) Feature Fusion Block and Output Block:

$$\begin{aligned} x_F &= \mathrm{up}(x_{\mathrm{inf}}, 2\times 2, 2) \\ x_{F_1} &= \mathrm{concate}(x_F, x_{\mathrm{new}}) \\ x_{F_2} &= \mathrm{ReLU}(\mathrm{BN}(\mathrm{conv}_{3\times 3}(x_{F_1}))) \\ x_{F_3} &= \mathrm{ReLU}(\mathrm{BN}(\mathrm{conv}_{3\times 3}(x_{F_2}))) \\ x_{F_4} &= \mathrm{BN}(\mathrm{conv}_{1\times 1}(x_{F_1})) \\ x_{F_5} &= x_{F_3} + x_{F_4}. \end{aligned} \tag{6}$$

The output block of MARNet is regressed on the feature map using a general convolutional structure to obtain two types of segmentation results: the area where *S. alterniflora* is located and the other land types.

B. Transfer Learning

MARNet applies two methods, i.e., MMD and direct entropy minimization, to optimize the loss function based on the segmentation loss of SD. These two methods optimize the segmentation of *S. alterniflora* in two directions, interdomain alignment and intradomain adaptation, respectively.

As shown in Table I, there is a serious data imbalance between *S. alterniflora* and the background. Since cross-entropy loss performs poorly in the face of extreme sample imbalance, MARNet uses the intersection over union (IoU) loss instead of cross-entropy loss when predicting SD

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probabilities and computing errors using labels. Assume the number of classification categories is C. The IoU loss of the labeled SD is defined as

$$\mathcal{L}_{\text{seg}}(X_s, Y_s) = -\frac{1}{C}\sum_{c=1}^{C}\frac{\sum_{i=1}^{n_s} y_i^c\, p_i^c}{\sum_{i=1}^{n_s}\left(y_i^c + p_i^c - y_i^c\, p_i^c\right)} \tag{7}$$

where y_i is the one-hot encoding of the label information and p_i is the predicted probability output obtained by the softmax function.

MARNet uses the optimization method of MMD in the direction of interdomain alignment and applies the MMD distance for the marginal distribution alignment between SD and TD. The spectral drift of the data is reduced in both domains from the perspective of aligning the statistical feature distributions of the source and TD data. In practice, the unbiased estimation of MMD is to compare the squared distance between the empirical kernel mean embeddings as

$$\begin{aligned}\mathcal{L}_{\text{MMD}}(Z_s, Z_t) = &\frac{1}{n_s^2}\sum_{i=1}^{n_s}\sum_{j=1}^{n_s} r\left(z_i^s, z_j^s\right) \\ &+ \frac{1}{n_t^2}\sum_{i=1}^{n_t}\sum_{j=1}^{n_t} r\left(z_i^t, z_j^t\right) - \frac{2}{n_s n_t}\sum_{i=1}^{n_s}\sum_{j=1}^{n_t} r\left(z_i^s, z_j^t\right)\end{aligned} \tag{8}$$

where the kernel $r(\cdot, \cdot)$ represents inner product of the features in RKHS.

In terms of intradomain adaptation, MARNet uses the optimization method of entropy minimization, introducing the entropy loss of TD to directly maximize its predictive certainty. The TD's feature map obtained from the output of the obtained network is input to the softmax layer to obtain the prediction probability map P_t, and then, the entropy map $E_t = \{e_i^t\}_{i=1}^{n_t} \in [0, 1]$ is obtained by applying the Shannon entropy [53]. The entropy loss $\mathcal{L}_{\text{ent}}$ is defined as the sum of all normalized entropy values

$$\begin{aligned} P_t &= \text{Softmax}(Z_t) \\ E_t &= \frac{-1}{\log(C)}\sum_{c=1}^{C} P_t^c \log P_t^c \\ \mathcal{L}_{\text{ent}} &= \sum_{i=1}^{n_t} e_i^t. \end{aligned} \tag{9}$$

For the TD, intradomain adaptation using the entropy minimization optimization can push the decision boundary of the model to the low-density region of the TD distribution in the prediction space. As shown in Fig. 3, this allows the results for predicting intergradation across years to have finer object edges and large blurred image regions to be correctly fixed.

Based on the above loss functions, MARNet combines supervised segmentation loss on source samples $\mathcal{L}_{\text{seg}}$, distribution alignment loss on the source and target samples $\mathcal{L}_{\text{MMD}}$, and unsupervised entropy loss on target samples $\mathcal{L}_{\text{ent}}$ during the training process, and the final optimization problem is formulated as follows:

$$\begin{aligned}\mathcal{L}(X_s, X_t, Y_s) = &\mathcal{L}_{\text{seg}}(X_s, Y_s) + \lambda_1 \mathcal{L}_{\text{MMD}}(Z_s, Z_t) \\ &+ \lambda_2 \mathcal{L}_{\text{ent}}(Z_t)\end{aligned} \tag{10}$$

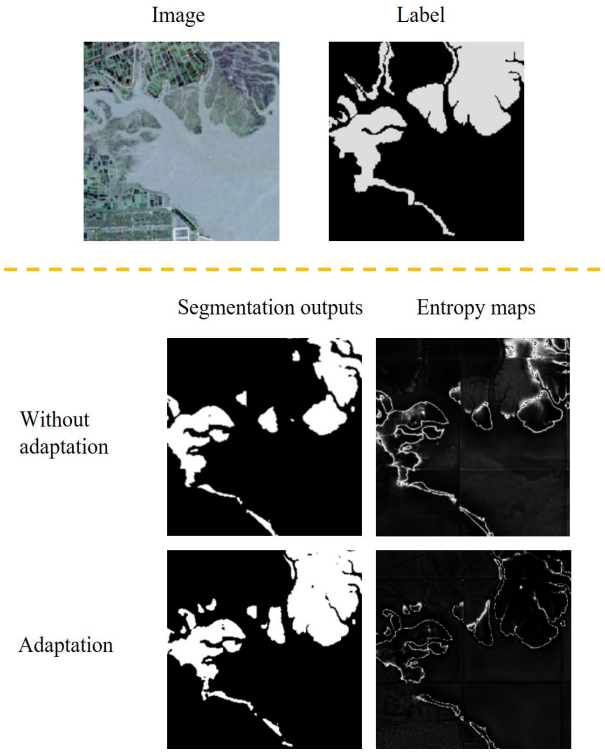


Fig. 3. Effect plots of the entropy minimization optimization method based on intradomain adaptation. The first row displays a section of the remote sensing image in TD and its corresponding true value map. The second and third rows illustrate the segmentation results of this remote sensing image before and after intradomain adaptation, along with their corresponding predicted entropy maps.

where λ_1 and λ_2 are the regularization parameters controlling the different contributions of interdomain alignment and intradomain adaptation. According to our empirical study and to reduce the pressure of hyperparameter optimization, we choose $\lambda_1 = 0.01$ and $\lambda_2 = 0.001$.

C. Redundant Cutting and Splicing

Redundant cutting and splicing is one of the post-processing approaches. For the predicted segmentation of *S. alterniflora*, the sample points are much larger in the background than in *S. alterniflora* due to the severe sample imbalance problem. Even the poor prediction of the boundary of some image blocks can lead to a very serious decrease in the overall index. Therefore, we adopt a post-processing approach to the prediction results of MARNet output.

The schematic of the post-processing method is shown in Fig. 4, which first fills the overall remote sensing image with the height and width of the fill equal to the height $(H/4)$ and the width $(W/4)$ of the cropping around in the post-processing, and then sets the horizontal step of the crop of the overall remote sensing image to $(H/2)$ and the vertical step to $(W/2)$. Then, the patches obtained from network prediction are cropped with corresponding width and height, and finally, the stitching of corresponding positions is performed.

IV. Experimental Results and Discussions

A. Dataset Description and Preprocessing

The dataset used in this study is distributed along the coastline of Jiangsu Province, China (between 119°21′E and

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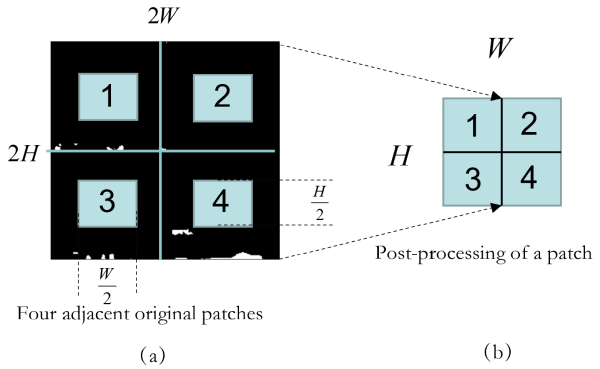


Fig. 4. Redundant cutting schematic. (a) Four adjacent patches were obtained by network prediction. (b) Single patch obtained by post-processing (a). The four neighboring patches are cut by redundant parts to obtain a single patch since they have predicted contents for the same position.

TABLE II
OVERVIEW OF LANDSAT DATA AFTER BAND COMBINATION USED IN THIS STUDY (DATE FORMAT: MONTH/DAY/YEAR)

Period	Sensor	Image Time	Resolution(m)	Number of Bands
1995	Landsat 5 TM	08/03/1995 08/12/1995 09/27/1995	30	6
2000	Landsat 5 TM	08/21/2000 09/17/2000 10/10/2000	30	6
2005	Landsat 5 TM	08/20/2004 10/08/2005 10/17/2005	30	6
2010	Landsat 5 TM	09/19/2009 10/03/2009 10/31/2010	30	6
2015	Landsat 8 OLI	08/03/2015 09/18/2015 10/13/2015	30	6
2020	Landsat 8 OLI	10/31/2019 08/16/2020 09/08/2020	30	6

121°55′*E*, 31°33′*N* and 35°07′*N*). The coastline of Jiangsu Province has a total length of 954 km. It stretches from the Xiuzhen estuary in the north to the estuary of the Yangtze River in the south and faces the Yellow Sea in the east. At present, Jiangsu coastal *S. alterniflora* lacks sufficient remote sensing interpretation data of historical years, which is very unfavorable for the scientific management of invasive and rapidly expanding *S. alterniflora*. In order to develop and protect coastal wetlands better, and to analyze the evolution of wetland geography in long time series more effectively, an experiment of intelligent interpretation of multitemporal remote sensing images is designed, i.e., the remote sensing data of unknown years are interpreted by the remote sensing data of existing years.

The dataset was constructed using a total of 21 original Landsat images for seven time periods: 1995, 2000, 2005, 2010, 2015, and 2020 (as listed in Table II). The selected remote sensing images were concentrated in August–October, when *S. alterniflora* was at its peak, and their spectral characteristics were well distinguished from other vegetation [54]. Due to the large study area and the fact that *S. alterniflora* is distributed only along the coastline, the inland part is cropped. It was extremely difficult to obtain images that completely cover the study area and meet the monthly conditions in one year in order to avoid the occlusion of the study area by clouds in the images. Therefore, three remote sensing images for each year were stitched together, and the missing images were obtained from adjacent years to reduce data loss due to cloud cover. Because the remote sensing data used in the study originated from different sensors with different spatial resolutions. In order to unify the data to a spatial resolution of 30 m, the missing 15-m panchromatic and multispectral bands of Landsat 5 TM in Landsat 8 OLI data were removed, and the six common bands of red, green, blue, NIR, SWIR-1, and SWIR-2 of Landsat 5 TM and Landsat 8 OLI were selected in this study. The original bands were selected. The wavelength ranges of NIR, SWIR-1, and SWIR-2 of Landsat 5 TM were 0.760–0.900, 1.550–1.750, and 2.080–2.350, respectively. The wavelength ranges of NIR, SWIR-1, and SWIR-2 of Landsat 8 OLI were 0.845–0.885, 1.560–1.660, and 2.100–2.300. In this article, a series of preprocessing of Landsat data was performed using ENVI 5.3 software. The digital numbers were converted to radiance by radiometric calibration. The atmospheric correction method used was quick atmospheric correction (QUAC). Finally, band combination, image mosaicking, and clipping were performed.

Due to the spectral shift caused by different years, different weather conditions, different sensors, and so on, the direct use of remote sensing data from other years to segment the remote sensing data of the target year will encounter the problem of spectral shift, which makes the final interpretation results less accurate. In order to improve the accuracy of the interpretation of remote sensing data of the target years, so as to achieve the monitoring of large areas and long time series of *S. alterniflora* in wetlands. Therefore, remote sensing data of other years were set as the SD, and remote sensing data of the target year were set as the TD. The experimental design aims to train the model using SD, transfer the knowledge shared in SD to TD, and segment TD to achieve the transfer of the model to cross-year scenarios.

The study takes remote sensing data of 2000, 2010, and 2020 as TD and the remaining years as SD to verify the validity of MARNet, Fig. 5 shows the remote sensing data and the corresponding labels for this three years. The years 2010 and 2020 are the last set of data for Landsat 5 TM and Landsat 8 OLI, respectively. Because the prediction of *S. alterniflora* in wetlands is needed, the years 2010 and 2020 are selected as TD and the remaining years as SD for two sets of experiments, respectively. Meanwhile, to confirm the validity of MARNet in the remaining years of Landsat 5 TM, the year 2000 in Landsat 5 TM is selected as the TD and the remaining years as the SD for the third set of experiments. Although cloud cover has been avoided as much as possible in 2015, there is still a large cloud cover in the nonstudy area of remote sensing data, as shown in Fig. 6, and there is no cloudy area in the rest of the years, so it is temporarily impossible to select 2015 as TD to verify the validity of MARNet in the rest of the years of Landsat 8 OLI.

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Fig. 5. Remote sensing images of different years and their corresponding labels. (a) 2000, (b) 2010, and (c) 2020.



Fig. 6. Overall area of the coastline and local areas covered by more clouds in 2015. The upper left corner (red boxed area) of the overall remote sensing image in 2015 is more severely covered by clouds, accounting for about one-third of the overall area.

The remote sensing images of each year and matching vector marked images are split, to obtain large numbers of patches of size 128 × 128 pixels. Due to post-processing, the split step for the year of SD is set to 64, and the split step for the year of TD is set to 128 and 64. Because three separate years of experiments are required, three different sets of SD and TD image pairs are obtained, respectively, with specific information as listed in Table III. Each image pair contains one remotely sensed image and one labeled image. *S. alterniflora* in the survey images is one category, and all other land types are classified as background.

B. Evaluation Metrics

The number of samples of *S. alterniflora* is too different from the number of background samples, so the IoU is chosen as the main evaluation metric of segmentation. In addition, because it is a binary classification problem, the segmentation effect is judged by using Precision and Recall of *S. alterniflora* class for prediction angle and true angle, respectively. Finally, the F1-score is used to scientifically and effectively measure each metric of the network in the binary classification. The computational efficiency is compared to the running time of a single round of network training.

The following indicators are often used to evaluate semantic segmentation methods. True positive (TP) is the pixel number that correctly predicts the target. False positive (FP) is the number of other categories of pixels that predict the target. False negative (FN) is the pixel number that predicts the target incorrectly. True negative (TN) indicates that other pixels are correctly predicted. The calculation methods of evaluation indicators are

$$\begin{cases} \text{IoU} & = \dfrac{\text{TP}}{\text{FN}+\text{FP}+\text{TP}} \\ \text{Precision} & = \dfrac{\text{TP}}{\text{TP}+\text{FP}} \\ \text{Recall} & = \dfrac{\text{TP}}{\text{TP}+\text{FN}} \\ \text{F1-score} & = \dfrac{2\text{TP}}{\text{FP}+2\text{TP}+\text{FN}} \end{cases} \tag{11}$$

C. Settings of Experiments

The experiments are carried out on a computer with a GeForce RTX 3090 GPU with 24 GB of RAM. The optimizer function is the stochastic gradient descent (SGD) function [55], and the learning rate is 0.03. The output of the network is a two-channel segmented prediction tensor. The value of each pixel's channel represents the probability that the pixel is *S. alterniflora* or background. The training epoch is set to 100. The batch size number is set to 8.

Attention-UNet [19], HRNet [27], DANet [41], CCNet [35], GCNet [34], Deeplab-v3 [26], PSPNet [28], ResNeSt [33], ISANet [42], ConvNeXt [30], ProCA [56], IUnet++ [57], and UNet-CNN [58] are selected for comparison with MARNet with only segmentation loss and no transfer learning loss. CBAM [40], CA [43], EPSA [44], PAB, and CAB [21] are selected for comparison with MA-block. The data used for training and testing of these networks are SD and TD without post-processing as listed in Table III, respectively.

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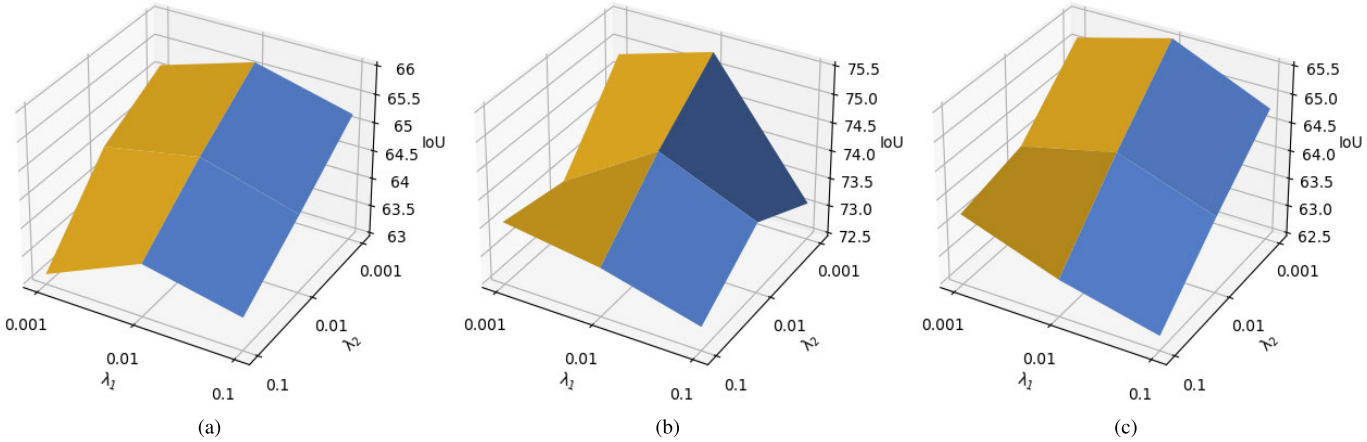


Fig. 7. Parameter tuning of λ_1 and λ_2 for the proposed MARNet using all three experimental data. (a) 2000, (b) 2010, and (c) 2020.

TABLE III
NUMBER OF IMAGES CORRESPONDING TO EACH EXPERIMENT

Year	Numbers of patches (steps=128)	Numbers of patches (steps=64)	Experiment1			Experiment2			Experiment3		
			Source	Target	Target (Post-processing)	Source	Target	Target (Post-processing)	Source	Target	Target (Post-processing)
1995	\	4669	✓			✓			✓		
2000	1164	4676	✓			✓				✓	✓
2005	\	4782	✓			✓			✓		
2010	1186	4734		✓	✓	✓			✓		
2015	\	4404	✓			✓			✓		
2020	1130	4525	✓				✓	✓	✓		
Sum			23065	1130	4525	23056	1186	4734	23114	1164	4676

The full MARNet (including post-processing) is also added for comparison with the MARNet. The SD and TD used in the full MARNet correspond to the source and target (post-processing) of the three experiments in Table III, respectively. The loss function is IoU loss, which is the same as the segmentation loss of MARNet. Since the number of source samples is n_s, and the number of classification categories is C. The value of C is 2, referring to the *S. alterniflora* and the background. Assuming that the true category of the sample is y_i^c, the predicted probability that sample i belongs to category c is p_i^c. The loss function used is

$$\text{Loss} = -\frac{1}{C}\sum_{c=1}^{C}\frac{\sum_{i=1}^{n_s} y_i^c\, p_i^c}{\sum_{i=1}^{n_s}\left(y_i^c + p_i^c - y_i^c\, p_i^c\right)}. \tag{12}$$

To ensure the accuracy of the experiment, an open-source semantic segmentation toolbox based on PyTorch called MMSegmentation [59] is used to complete the comparison experiments, except for MARNet. To ensure fairness, all the networks are not preloaded with training parameters. The batch size is set to 8, and the same segmentation loss function is used. The experimental results of each network are the average of five executions.

In the proposed MARNet, the adjustable hyperparameters are the regularization parameters, namely λ_1 and λ_2. The optimal parameters are selected using tenfold cross-validation, with a range of 1e-3, 1e-2, and 1e-1 for the regularization parameters. λ_1 and λ_2 are crucial for MARNet as they control the alignment of *S. alterniflora* segmentation distribution in multitemporal remote sensing data and the network's ability to make high-confidence predictions. Fig. 7 shows the trend of MARNet's segmentation performance with different variations in the three testing years' data. To demonstrate the model's generalization ability on different testing data and reduce the pressure of hyperparameter tuning, λ_1 and λ_2 are set to 0.01 and 0.001, respectively, for all three testing years' data.

D. Segmentation Performance

MARNet, which contains only segmentation loss, is trained with other contrast semantic segmentation networks using SD. The inclusion of segmentation loss only means that the loss function of MARNet is the same as other comparative semantic segmentation networks, without the loss function of transfer learning and post-processing. Both SD and TD (post-process) are used to train the overall MARNet, as transfer learning requires the use of TD remote sensing data. Each network is trained five times, and the segmentation scores are averaged to ensure the stability of the model.

Table IV lists the test results of the segmentation networks for 2010 remote sensing images as target scenes. Remote sensing data for this year are provided by Landsat 5 TM. The experiments demonstrate that MARNet has the highest IoU with an IoU score of 72.83 for all networks with only

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TABLE IV
COMPARISON OF THE EVALUATION METRIC SCORES (%) AMONG DIFFERENT SEGMENTATION NETWORKS IN 2010

Network	Experimental Results			
	IoU	Precision	Recall	F1-Score
Attention-UNet [19]	69.53	86.75	77.79	81.99
HRNet [27]	67.18	80.52	80.24	80.25
DANet [41]	66.93	82.32	78.20	80.18
CCNet [35]	66.15	80	79.25	79.61
GCNet [34]	66.58	81.96	78	79.93
Deeplab-v3 [26]	66.6	79.42	80.55	80.25
PSPNet [28]	66.45	79.28	80.51	81.77
ResNeSt [33]	66.53	81.6	78.51	79.89
ISANet [42]	64.87	87.39	71.66	78.68
ProCA [56]	66.47	86.73	74.09	79.94
ConvNeXt [30]	53.29	73.27	66.15	69.53
IUNet++ [57]	59.51	83.8	67.24	74.62
UNet-CNN [58]	65.41	84.71	74.16	79.08
MARNet(only seg)	72.83	88.35	80.77	84.29
MARNet(all)	**77.54**	**89.41**	**85.38**	**87.34**

TABLE V
COMPARISON OF THE EVALUATION METRIC SCORES (%) AMONG DIFFERENT SEGMENTATION NETWORKS IN 2000

Network	Experimental Results			
	IoU	Precision	Recall	F1-Score
Attention-UNet [19]	62	68.82	86.28	76.52
HRNet [27]	62.16	71.04	83.57	76.66
DANet [41]	61.42	73.53	78.88	76.09
CCNet [35]	62.46	74.11	79.96	76.89
GCNet [34]	60.65	73.58	80.24	75.51
Deeplab-v3 [26]	63.24	73.36	82.13	77.47
PSPNet [28]	63.26	71.72	84.33	77.48
ResNeSt [33]	60.66	71.58	80.32	75.5
ISANet [42]	61.05	77.9	74.09	75.8
ProCA [56]	60.59	79.92	71.46	75.4
ConvNeXt [30]	61.99	73.83	79.54	76.52
IUNet++ [57]	56.09	64.86	80.58	71.87
UNet-CNN [58]	60.86	71.27	80.64	75.67
MARNet(only seg)	64.49	72.96	84.79	78.39
MARNet(all)	**67.07**	**75.56**	**85.95**	**80.29**

segmentation loss included. After adding the loss function for post-processing and the loss function for transfer learning, the IoU scores of MARNet can reach 77.54. Table IV indicates that the segmentation results of MARNet are also better than the other networks. Attention-Net, HRNet, DANet, and DeepLab-v3 have general segmentation results, with IoU scores exceeding 66.6. Attention-Net is one level below MARNet in the results of segmentation and has the second-best performance on several metrics. The IoU score of MARNet is 3.3 higher than that of Attention-UNet with the same loss function, and the scores of other metrics are also higher than that of Attention-UNet. This demonstrates that MARNet can learn the *S. alterniflora* category better than other networks by the network framework alone and extract the features of *S. alterniflora* well. Meanwhile, the performance is further improved with the addition of post-processing and transfer learning components. Table V lists the test results of the segmentation networks for the 2000 remote sensing image as the target scene, also provided by Landsat 5 TM. This year, the four comparison networks with better metrics differed from 2010, which are PSPNet, DeepLabv3, CCNet, and HRNet. but MARNet is still the best among all networks. Combining the two sets of experiments, MARNet outperforms other networks in the segmentation of *S. alterniflora* even under different years.

TABLE VI
COMPARISON OF THE EVALUATION METRIC SCORES (%) AMONG DIFFERENT SEGMENTATION NETWORKS IN 2020

Network	Experimental Results			
	IoU	Precision	Recall	F1-Score
Attention-UNet [19]	60.92	76.54	74.88	75.7
HRNet [27]	59.87	75.75	74.21	74.89
DANet [41]	59.88	76.06	73.91	74.9
CCNet [35]	60.98	76.3	75.26	75.75
GCNet [34]	61.21	75.79	76.38	75.93
Deeplab-v3 [26]	61.8	77.18	75.66	76.38
PSPNet [28]	62.01	77.6	75.57	76.55
ResNeSt [33]	55.5	74.23	68.77	71.38
ISANet [42]	60.57	80.36	71.17	75.4
ProCA [56]	57.44	72	73.77	72.89
ConvNeXt [30]	51.01	65.83	69.37	67.56
IUNet++ [57]	56.35	75.98	68.57	72.09
UNet-CNN [58]	58.32	76.23	71.28	73.67
MARNet(only seg)	65.12	77.98	79.87	78.89
MARNet(all)	**66.8**	**80.94**	**79.3**	**80.08**

Table VI lists the test results of the segmentation networks for 2020 remote sensing images as target scenes. The remote sensing data for this year are provided by Landsat 8 OLI. In addition to MARNet, PSPNet, DeepLabv3, GCNet, and CCNet perform better. However, with only segmentation loss, MARNet improves IoU by 3.11, Precision and Recall by 0.38 and 4.3, respectively, and F1-score by 2.34 relative to PSPNet, which has the highest metrics in the comparison network. With the addition of post-processing and transfer learning components, MARNet's performance is further improved. Since the sources of remote sensing data in 2020 are different from those in 2010 and 2000.

Figs. 8–10 illustrate the segmentation results of MARNet and several other networks for the remote sensing data of 2000, 2010, and 2020. Remote sensing data from Landsat 5 TM for 2000 and 2010, and from Landsat 8 OLI for 2020, contain some typical *S. alterniflora* distribution areas in different experimental areas. Because the same land is different from year to year and the spectral shift is caused by different sensors, different segmentation effects exist for the same network. Therefore, the comparison network with the top four combined indicators in the three sets of experiments is selected for the comparison of segmentation results.

Again, MARNet has the best segmentation performance. The other networks have different focuses on feature learning for *S. alterniflora*, so the segmentation results differ slightly. From Figs. 8 and 9, MARNet has the best segmentation results even for remote sensing data of different years. For images with more *S. alterniflora*, such as the second and third rows

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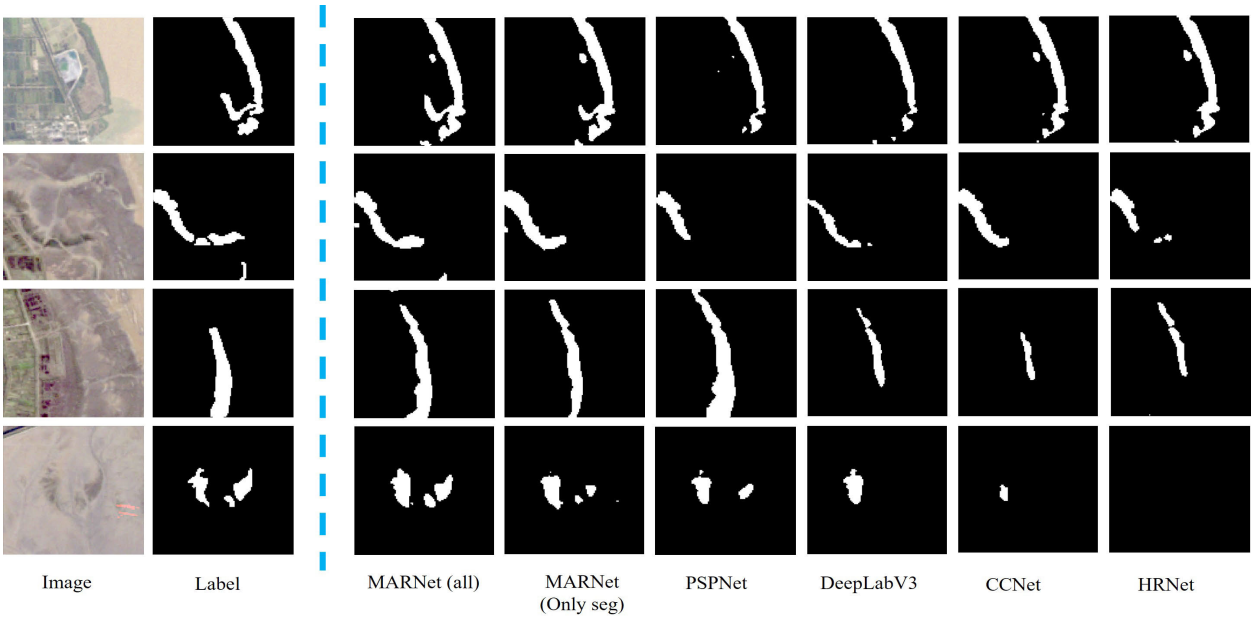


Fig. 8. Segmentation results of remote sensing data in 2000 by MARNet and other networks. The remote sensing data source for this year is Landsat 5 TM. These results show the local comparison of the prediction results of MARNet and some networks for remote sensing images in 2000.

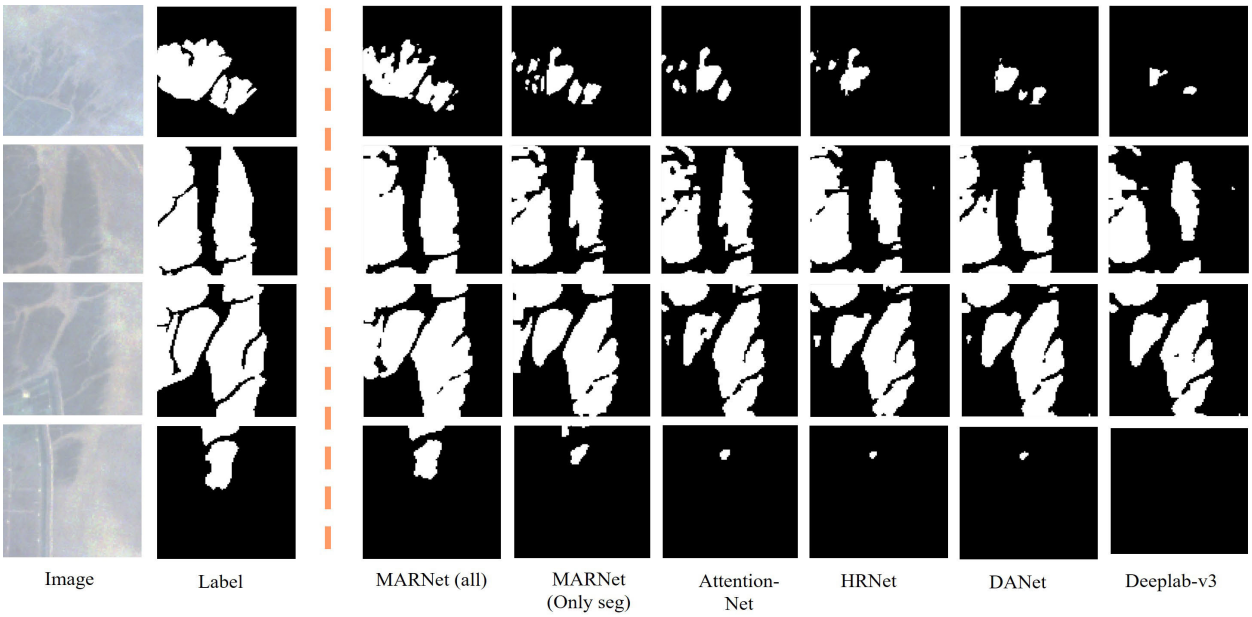


Fig. 9. Segmentation results of remote sensing data in 2010 by MARNet and other networks. The remote sensing data source for this year is Landsat 5 TM.

of Fig. 9, MARNet also shows better performance than other networks. Meanwhile, with the addition of post-processing and transfer learning components, MARNet's extraction ability is further enhanced for *S. alterniflora* with smaller targets. In summary, MARNet has the best segmentation performance for *S. alterniflora* with different years and different land types. For remote sensing data with different sensors and different years, MARNet is best in detail processing and overall segmentation compared with other comparison networks. This excellent segmentation comes from MARNet's attention structures, transfer learning loss functions, and post-processing.

Tables VII–IX present the test results of a segmentation network on remote sensing data from 2000, 2010, and 2020, with and without the addition of various attentional modules including the MA-block. The results demonstrate that the MA-block not only improves the recognition of *S. alterniflora* but also shows higher computational efficiency than other advanced attention structures.

Based on the comparison experiments in Tables V, VI, and X, MARNet with only segmentation loss achieved the highest computational efficiency and best segmentation performance in the three-year comparison. Although MARNet (all) with domain alignment and adaptation losses had a longer computation time and some loss in computational efficiency, the overall segmentation performance was greatly improved, which is crucial for the detection and prevention of the invasive plant *S. alterniflora*.

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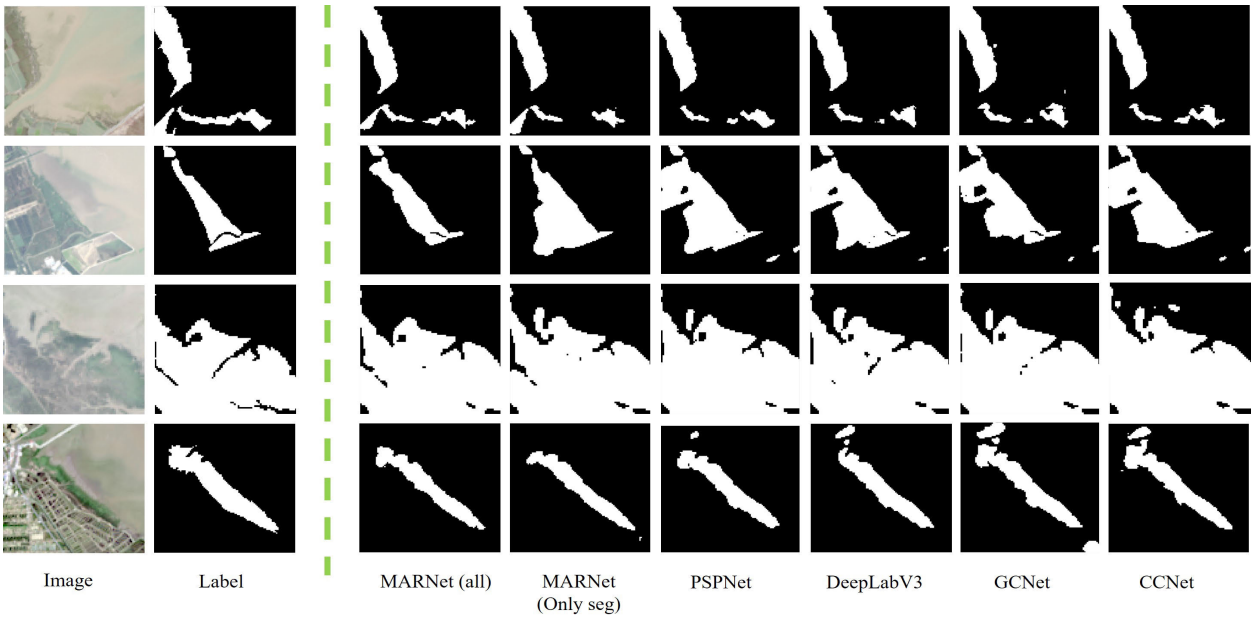


Fig. 10. Segmentation results of remote sensing data in 2020 by MARNet and other networks. The remote sensing data source for this year is Landsat 8 OLI.

TABLE VII
COMPARISON OF THE EVALUATION METRIC SCORES (%) AMONG DIFFERENT AMS IN 2000

Attention Mechanism	CBAM	CA	EPSA	PAB+CAB	MA
IoU	61.38	61.66	62.17	60.5	**64.49**
Precision	68.68	68.42	69.33	67.4	**72.96**
Recall	85.48	86.48	85.83	85.7	**84.79**
F1-Score	76.04	76.28	76.67	75.4	**78.39**
TrainTime (s)	179.95	168.39	296.65	165.45	**164.91**
TestTime (s)	3.55	3.172	4.98	3.2	**3.09**

TABLE VIII
COMPARISON OF THE EVALUATION METRIC SCORES (%) AMONG DIFFERENT AMS IN 2010

Attention Mechanism	CBAM	CA	EPSA	PAB+CAB	MA
IoU	66.78	67.96	69.18	68.37	**72.83**
Precision	85.62	**90.68**	89.46	84.5	88.35
Recall	75.24	72.88	75.44	78.43	**80.77**
F1-Score	80	80.74	81.76	81.23	**84.29**
TrainTime (s)	178.62	166.59	291.21	165.28	**163.96**
TestTime (s)	3.644	3.23	4.84	3.27	**3.11**

V. ABLATION STUDY

In order to verify the effectiveness of each block, this section progressively adds each structure to the base network and performs ablation experiments. The base network of MARNet is the U-Net with the addition of residual connections. The results of the experiments are listed in Table XI.

A. Ablation for the MA-Block

The first row (R0) and the second row (R1) of Table XI demonstrate that the MA-block enables MARNet to focus on features in a different way, improving the ability to extract features from the *S. alterniflora* category. The effectiveness of

TABLE IX
COMPARISON OF THE EVALUATION METRIC SCORES (%) AMONG DIFFERENT AMS IN 2020

Attention Mechanism	CBAM	CA	EPSA	PAB+CAB	MA
IoU	62.55	61.33	62.58	61.3	**65.12**
Precision	78.32	75.27	78.78	**79.56**	77.98
Recall	75.66	76.83	75.26	72.82	**79.87**
F1-Score	77.01	76.03	76.94	75.98	**79.89**
TrainTime (s)	159.98	157.38	282.67	157.2	**155.28**
TestTime (s)	3.12	3.01	4.92	3.12	**2.9**

TABLE X
COMPARISON OF SINGLE-EPOCH RUN TIMES (S) AMONG DIFFERENT NETWORKS FOR ALL YEARS

Network	TrainTime			AverageTime
	2000	2010	2020	
Attention-UNet [19]	191.19	190.2	189.21	190.2
HRNet [27]	421.26	415.35	411.24	415.95
DANet [41]	169.61	167.38	165.32	167.44
CCNet [35]	167.91	166.69	162.21	165.61
GCNet [34]	166.08	**163.05**	157.11	162.08
Deeplab-v3 [26]	165.04	162.33	161.49	162.95
PSPNet [28]	167.76	165.39	162.76	165.31
ResNeSt [33]	305.09	304.06	296.44	301.86
ISANet [42]	174.37	172.74	167.87	171.66
ProCA [56]	393.21	387.94	377.58	171.66
ConvNeXt [30]	246.11	242.02	239.97	242.7
IUNet++ [57]	312.61	308.84	304.41	308.62
UNet-CNN [58]	206.42	200.21	191.83	199.49
MARNet(only seg)	**164.91**	163.96	**155.28**	**161.38**
MARNet(all)	287.15	286.92	285.49	286.52

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TABLE XI
SCORE (%) OF MARNET AFTER ABLATION OF EACH STRUCTURE

	Model				Experimental Results (2000)				Experimental Results (2010)				Experimental Results (2020)			
	MA	MMD	ENT	Post-processing	IoU	Precision	Recall	F1-Score	IoU	Precision	Recall	F1-Score	IoU	Precision	Recall	F1-Score
R0					63.84	70.86	86.4	77.84	70.78	89.52	75.16	81.64	63.86	79.8	75.96	77.84
R1	✓				64.49	72.96	84.79	78.39	72.83	88.35	80.77	84.29	65.12	77.98	79.87	78.89
R2	✓	✓			65.46	73.11	86.35	79.13	73.94	88.78	81.6	85.02	65.24	79.64	78.68	79
R3	✓		✓		65.2	71.46	88.2	78.94	74.82	89.58	82	85.6	65.32	78.68	79.6	79.02
R4	✓	✓	✓		65.64	73.88	85.63	79.24	75.32	89.66	82.56	85.92	65.56	79.6	78.86	79.2
R5	✓	✓	✓	✓	67.07	75.56	85.95	80.29	77.54	89.41	85.38	87.34	66.8	80.94	79.3	80.08

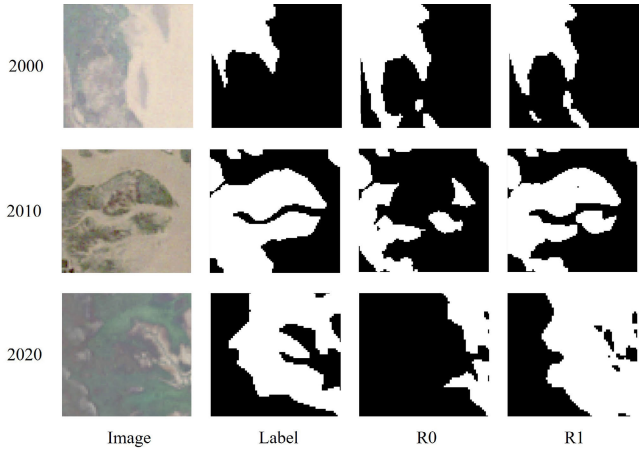


Fig. 11. Results of various ablation experiments on the MA-block. R0 indicates the segmentation effect of the network frame without any optimization on *S. alterniflora*, and R1 indicates the segmentation effect on *S. alterniflora* with the addition of an MA-block to the network frame.

the MA-block is confirmed by three sets of experiments in 2000, 2010, and 2020. Compared with the basic network with only residual connections added, the addition of MA-block can improve IoU by 0.65, 2.05, and 1.26, respectively, and the precision and other indexes of *S. alterniflora* class are improved to different degrees in the three-year ablation experiments. Meanwhile, the experimental results in Section IV demonstrate that the network containing only the segmentation loss has reached a relatively good segmentation level after the addition of MA-block.

As shown in Fig. 11, the segmentation results are improved using MA-block compared with the original network. The base network does not extract the semantic information of *S. alterniflora* well, while the semantic divide is large. However, using MA-block, the network combined the distribution characteristics and spectral characteristics of *S. alterniflora* to extract more features from three directions: channel attention, contextual attention, and spatial attention, which enhanced the transmission of semantic information.

B. Ablation for the Transfer Learning

R2 and R1 of Table XI demonstrate that adding an interdomain alignment loss function to reduce the interdomain gap between SD and TD improves MARNet's learning ability of *S. alterniflora* category features in TD. MMDLoss enables the network to reduce the spectral shift in terms of aligning the distribution of SD and TD statistical properties, and the domain-invariant features learned from it alleviate the spectral shift problem due to the different years. IoU and other metrics of *S. alterniflora* are improved by the addition of MMDLoss.

As shown in Fig. 12 for R1 and R2, the segmentation capability of the network is further improved by adding MMDLoss. Although the remote sensing data of the adjacent year in the SD, i.e., 2015, has the problem of large cloud coverage and more serious data loss. MMDLoss can still slightly improve the segmentation effect of 2020 by reducing the spectral shift problem.

R3 and R1 of Table XI demonstrate that adding an in-domain adaptive loss function to reduce the network's high entropy prediction of target-like images improves MARNet's ability to segment the category of *S. alterniflora* in TD. ENTLoss constrains MARNet by minimizing the entropy distribution of TD, which optimizes the edges of the otherwise blurrier images. As shown in Fig. 12 for R3 in 2020, the addition of ENTLoss distinguishes the original river from *S. alterniflora* compared with R1. The comparison between R3 and R1 in 2000 and 2010 that ENTLoss improves the overall partitioning ability of the network.

In summary, adding domain alignment and domain adaptation loss functions separately, as shown in Fig. 12 and three sets of ablation experiments R1–R3, improves the segmentation results of the network for *S. alterniflora* compared with the base network with only the MA-block added. Table XI of R4 and R1 demonstrates that adding both domain alignment and domain adaptation loss functions to MARNet can further improve the segmentation ability of the network in detail.

C. Ablation for the Post-Processing

R4 and R5 of Table XI demonstrate that after adding the redundant cut post-processing module to MARNet, there is a large improvement in all metrics in the ablation experiments for the three different sets of years. MARNet requires a post-processing component, which is due to the fact that the number of sample points in the background class is much larger than that of *S. alterniflora*, even a small number of background image blocks with poor edge prediction can lead to a large drop in the overall metrics.

Therefore, after adding the post-processing part to MARNet, the stability of the network can be enhanced while the accuracy of the network is guaranteed, and the overall prediction effect of the network on remote sensing images can be further improved. As shown in Fig. 13, after adding

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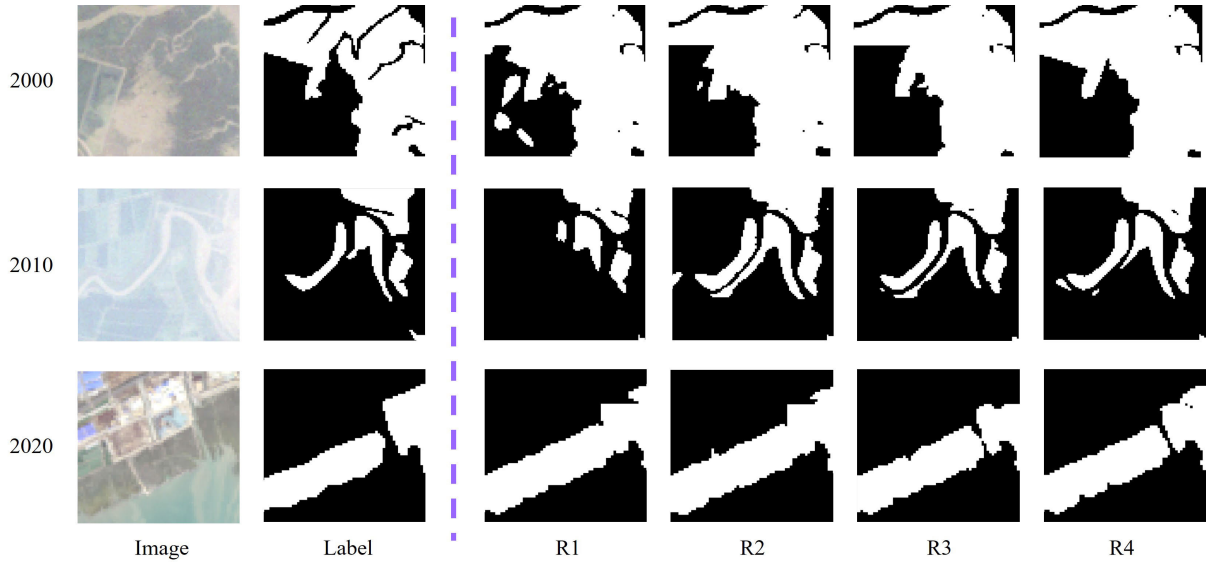


Fig. 12. Results of various ablation experiments with transfer learning loss functions. R1 denotes the segmentation effect on *S. alterniflora* by adding only MA-block to the original network framework, R2 denotes the segmentation effect on *S. alterniflora* by adding MA-block and MMDLoss to the original network framework, R3 denotes the segmentation effect on *S. alterniflora* by adding MA-block and ENTLoss to the original network framework, and R4 denotes the segmentation effect on *S. alterniflora* by adding MA-block, MMDLoss, and ENTLoss to the original network framework.

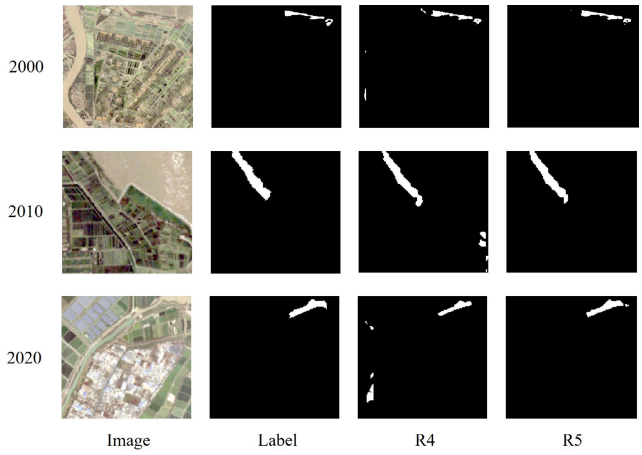


Fig. 13. Results of various ablation experiments of post-processing. R4 and R5 indicate the segmentation effects on *S. alterniflora* without and with the addition of post-processing in the network framework, respectively.

the post-processing part, the parts of the image block with small prediction errors appearing at the edges are cropped off, reducing the number of misclassified pixels and ultimately improving the network's recognition ability for *S. alterniflora*.

VI. Conclusion

In this article, a framework called MARNet is proposed for the *S. alterniflora* segmentation of multispectral remote sensing images. MARNet uses MA-block to identify *S. alterniflora* from various features of wetland vegetation. The block is designed according to the spectral characteristics and distribution properties of *S. alterniflora*, which enables MARNet to better feature extraction of *S. alterniflora* from multispectral remote sensing images. Meanwhile, MARNet achieves distribution alignment and model constraints from both interdomain alignment and intradomain adaptation perspectives through MMD and ENT to enhance the network segmentation. In addition, MARNet improves the overall prediction of remote-sensing images by the network through a post-processing approach. In a comprehensive experiment of predicting *S. alterniflora* across years on multispectral remote sensing images from three different years, the model is verified to have better performance compared with other deep learning-based semantic segmentation methods.

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